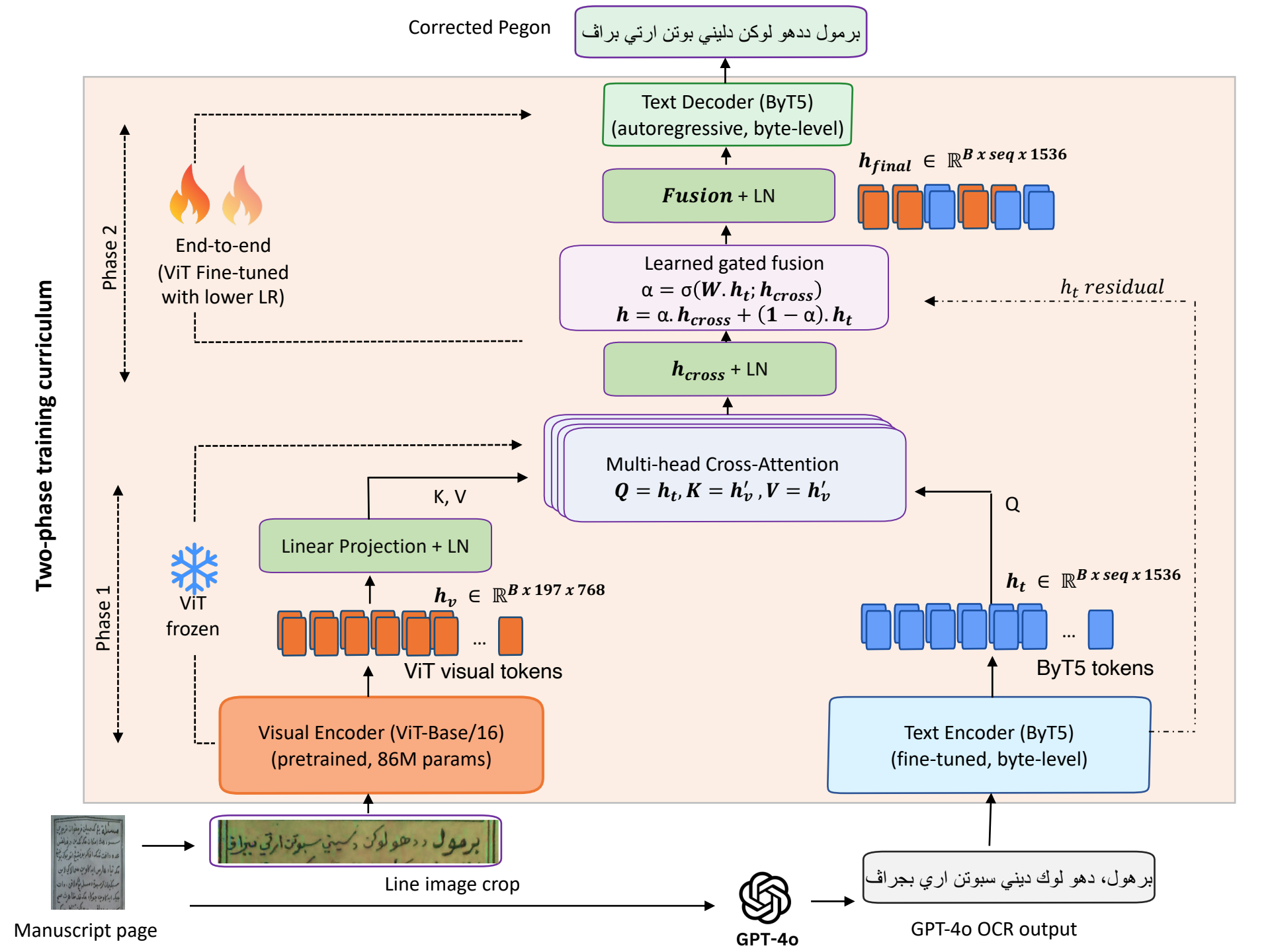

Corrected Pegon
برمول ددهو لوكن دليني بوتن ارتي براڤ
Text Decoder (ByT5)
(autoregressive, byte-level)
$h_{final} \in \mathbb{R}^{B\ x\ seq\ x\ 1536}$
Fusion + LN
Learned gated fusion
$\alpha = \sigma(W.h_t; h_{cross})$
$h = \alpha.h_{cross} + (1-\alpha).h_t$
h_t residual
h_{cross} + LN
Multi-head Cross-Attention
$Q = h_t, K = h'_v, V = h'_v$
K, V
Q
Linear Projection + LN
$h_v \in \mathbb{R}^{B\ x\ 197\ x\ 768}$
$h_t \in \mathbb{R}^{B\ x\ seq\ x\ 1536}$
ViT visual tokens
ByT5 tokens
Visual Encoder (ViT-Base/16)
(pretrained, 86M params)
Text Encoder (ByT5)
(fine-tuned, byte-level)
Phase 2
End-to-end
(ViT Fine-tuned
with lower LR)
Phase 1
ViT
frozen
Two-phase training curriculum
Line image crop
Manuscript page
GPT-4o
برهول، دهو لوك ديني سبوتن اري بجراڤ
GPT-4o OCR output